An explainable algorithm for credit risk prediction based on contrast patterns

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Abstract. Resumen

Keywords: credit risk · pattern recognition · clustering.

1 Introduction

Credit risk prediction involves determining whether an applicant will default on a future financial obligation [1], and is therefore a fundamental task in the financial sector. In the literature, this topic is often referred to using different terms such as credit rating, credit scoring, loan evaluation, credit bond, and creditworthiness [2–4]. In this work, we will use credit risk prediction. Furthermore, the problem is formulated as a classification task, where the objective is to assign each applicant to a class (e.g., default or non-default), rather than estimating a continuous risk value. It directly impacts the stability of banks and financial institutions, where accurate and explainable predictions are necessary to ensure regulatory compliance [5, 6]. Increasingly, financial institutions have implemented evaluations and management processes that employ pattern recognition and machine learning algorithms for credit risk prediction [2, 7]. As these predictors become more widely adopted, improving default prediction enables more accurate credit risk prediction and reduces losses [3, 4, 8].

Recent research in credit risk prediction has prioritized improving the performance of credit risk prediction algorithms based on machine learning and pattern recognition through the use of oversampling or undersampling methods, thereby reducing algorithmic bias towards the majority class [9–13]; the use of feature selection methods for identifying the most relevant features of a dataset to improve the performance of prediction algorithms [1, 14–16]; and the use of explainable AI techniques [10, 17–19] that allow users to comprehend and trust the predictions provided by credit risk prediction algorithms, which makes the predictions of the algorithm to comply with regulations.

Machine learning algorithms have demonstrated good predictive performance, and decision trees are among the most effective for credit risk prediction [7, 20–

25]. However, the need for explanations remains a significant limitation, particularly in the financial sector, as financial institutions and regulators require justification for their decisions [26, 27]. Several authors have addressed the lack of explainability in credit risk prediction [17, 28, 29].

Explainability techniques, also known as XAI (eXplainable Artificial Intelligence), aim to make machine learning prediction processes transparent and understandable to humans. These techniques are important in areas such as credit risk prediction, where justifying and explaining the predictions is essential in enabling financial institutions to make more informed decisions. Several recent studies have applied SHAP [30] and LIME [31] to credit risk prediction to generate explanations for individual predictions [32–34]. SHAP assigns to each feature a value that represents how much it contributed to the prediction. These values help explain the role each feature played in the prediction. LIME builds a simple interpretable predictor that approximates the behaviour of the original predictor by modifying a specific instance and observing the changes in the prediction. The result is an explanation that highlights the features contributing most to the prediction, showing how each affects the prediction in a positive or negative direction. However, these techniques focus on individual features and often overlook how several features jointly contribute to the prediction.

In credit risk prediction, the prediction produced by a predictor depends on multiple features rather than on individual features. Therefore, predictors should consider more than one feature to better explain the reasons behind the prediction [18, 35]. In response to this need, this article proposes a credit risk prediction algorithm that generates explanations based on contrast patterns, which are properties of the form $feature\#value$ that are characteristic of one class but rare in the opposite class. Because these patterns are efficiently mined with decision-tree-based miners, we propose using decision-tree-based predictors to predict credit risk, as the literature reports that they yield competitive credit risk predictions compared to the most successful ones. In this way, the proposed algorithm leverages the affinity and synergy between these two decision-tree-based algorithms.

Unlike other explainable algorithms that merely highlight the importance of individual features, our method explains predictions based on contrast patterns present in the training data and shared with the new applicant. Experiments conducted on both public and private financial datasets show that the proposed algorithm achieves competitive predictive accuracy while offering instance-level explanations based on the contrast patterns shared with other clients in the class, unlike the current state-of-the-art, which explains based on the single features contributing most to the prediction.

The remainder of this paper is organised as follows. Section 2 presents the related work. Section 3 describes the proposed algorithm. Section 4 presents our

experimental results and a discussion about them. Finally, Section 5 presents our conclusions and future work directions.

2 Related works

Credit risk prediction has been extensively studied in the literature using a wide range of pattern recognition and machine learning algorithms [4, 6, 8, 10, 14, 15, 28, 29, 33, 36–48]. In this section, we review the most recent works on credit risk prediction, with a focus on approaches that either prioritize achieving the highest accurate credit risk prediction or incorporate explainability for credit risk prediction. The works considered in this review are those that utilize publicly available datasets, commonly used in the literature and accessible online, which allows a comparison against our proposal. Accordingly, this section is divided into two parts: works that focus solely on accuracy without offering explanations, and works that integrate explainability into their predictors.

2.1 Predictors for Credit Risk Prediction

Recent works on credit risk prediction have focused on developing predictive algorithms to improve performance metrics, such as accuracy, area under the curve (AUC), Type II error, or F1-score.

In [38], the authors proposed a stacked ensemble for credit risk prediction that combines Random Forests (RF), Logistic Regression (LR), and Convolutional Neural Networks (CNN). The results of these predictors are used as input features for a Multilayer Perceptron (MLP), which learns to combine them to produce the prediction. The authors used SMOTEENN [49] to address class imbalance, and the stacked ensemble is compared against CNN, LR, MLP, and RF. After applying SMOTE-ENN to address class imbalance, the stacked ensemble consistently outperformed the individual predictors, achieving accuracies near 96%.

A comparative study of different predictors for credit risk prediction is presented in [46], including LR, RF, and Artificial Neural Networks (ANN). In this study, SMOTE [50] was employed to balance the training dataset, leading to significant improvements in the performance metrics of all predictors. Among them, RF achieved the best results, with an accuracy of 98%.

In [47], Deep Convolutional Neural Networks (DCNN) and Support Vector Regression (SVR) were compared for credit risk prediction. The application of SMOTE contributed to improved predictive performance in both predictors. Using SMOTE to balance the dataset, SVR achieved an accuracy of 98% and F1-score of 87%.

In [51], the authors present the Level-Wise Feature-Guided Cascading Ensembles (LFGCE) model, which integrates hierarchical feature selection with a cascading ensemble learning scheme. To evaluate its performance, several predictors were employed, including RF, Support Vector Machines (SVM), LR, Decision Trees (DT), ANN, Linear Discriminant Analysis (LDA), and XGBoost. Among these, XGBoost achieved the best results in both feature selection and prediction, with an AUC of 94%. However, it is important to note that the study does not address class imbalance, as no data balancing methods were applied.

Most of the reviewed studies emphasise improving predictive performance through class balancing methods to mitigate data imbalance. Several predictors yield good results in credit risk prediction. However, the reported performance metrics (such as accuracy, AUC, and F1-score) vary significantly on the chosen predictors and balancing methods. Overall, the authors consistently point out that addressing class imbalance is crucial for achieving reliable predictions, with many studies reporting high accuracy values, often exceeding 90%.

2.2 Explainable Predictors for Credit Risk Prediction

Incorporating explainability into credit risk prediction has been a growing area of study. These studies aim to achieve robust predictive performance while prioritising the explainability of predictions.

A study by [43] addresses credit risk prediction by comparing the predictive performance of XGBoost, RF, Least Squares SVM (LSSVM), and a CNN. After applying SMOTE to balance the dataset, XGBoost achieved the best predictive accuracy, reaching 92%. To enhance predictor explainability, SHAP analysis was applied to quantify the individual contributions of each feature. From the predictors evaluated, XGBoost performed best in terms of predictive accuracy and also provided explanations through SHAP values.

In [33], the authors present an ensemble framework that combines RF and XGBoost using probability-weighted averaging. The framework was evaluated using eight predictors, including LR, DT, Gradient Boosting (GB), LightGBM (LGBM), MLP, Naive Bayes (NB), and XGBoost. Their proposed ensemble significantly outperformed all others, achieving an AUC of 97%. SHAP was employed to explain the predictions, which allowed them to determine the contribution of each feature.

In [29], XGBoost is used as a predictor, combined with SHAP, LIME, and K-means to enhance explainability, achieving an AUC of 78%. SHAP and LIME are used to generate feature-contribution vectors for each training instance, which are grouped using K-means. For explaining the prediction of a new instance, its feature-contribution vector is calculated and assigned to the nearest cluster, and the prediction explanation is provided as the cluster's average vector.

In [6], the authors present a comparative analysis of the performance of boosting algorithms, including LightGBM, XGBoost, CatBoost, and AdaBoost. Additionally, SMOTE-Tomek [52] is employed to balance the dataset. The results indicate that LightGBM achieves the best performance with 95% accuracy and 98% AUC, outperforming the other boosting methods. SHAP is applied to quantify the contribution of each feature to the prediction.

In [28], a multilayer dynamic graph neural network (DYMGNN), proposed for credit risk prediction. The authors compare their approach against several baselines: LR, XGBoost, and DNN. Unlike many related studies, no resampling or class balancing methods are applied. Experimental results show that the DYMGNN achieves the best results with 81% accuracy and 83% F1-score. For explainability, SHAP is used to identify the features that contribute most to the predictions.

In [48], the authors propose an explainable two-stage technique to explain the results of credit risk prediction. In the first stage, SHAP is applied to analyse the importance of features according to their positive or negative impact on the prediction. In the second stage, they use a genetic algorithm to generate counterfactual explanations by perturbing the features according to this categorisation, thereby identifying the minimum necessary changes to modify the prediction. This technique was evaluated using multiple predictors, including SVM, XGBoost, MLP, and LSTM, with an accuracy of 90% for the employed predictors. The results demonstrate that the features modified most frequently closely align with the features identified as most relevant by SHAP.

In [53], the authors compare the performance of several Deep Learning (DL) architectures for credit risk prediction, including Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRU), and Graph Neural Networks (GNN). To address class imbalance, they apply SMOTEENN. The experimental results show that GRU outperformed all other predictors, achieving an accuracy of 91%. To explain the prediction, SHAP is used to quantify the importance of features and their contributions to the predictions.

In [8], the authors propose an approach for credit risk prediction on imbalanced datasets by integrating the TabTransformer model with a weighted loss function. This approach helps mitigate the bias toward the majority class, improving predictive performance on minority classes, achieving 91% accuracy and an AUC of 93%. SHAP is employed to evaluate feature importance and identify those features that contribute most to the predictions.

In [54], the authors propose a robust framework for credit risk prediction on imbalanced datasets by introducing the K-SMOTEENN method, which integrates oversampling and noise-filtering methods with a stacking ensemble. The

ensemble combines Random Forest, XGBoost, and LightGBM, resulting in a predictive performance of 92% F1-score, and 96% AUC. SHAP is used to analyse and explain the contributions of individual features to the prediction.

As discussed in the above works, credit risk prediction has been addressed through a wide range of predictors, balancing methods, feature selection methods, and evaluated through multiple performance metrics. Nevertheless, accurately detecting defaults remains a significant challenge, as no single method has consistently outperformed others across different scenarios. Moreover, the most commonly used explainability approaches, such as SHAP and LIME, while helpful in assessing the individual contribution of features, fail to capture feature value combinations that characterise different types of applicants. This limitation is particularly relevant, since financial institutions increasingly demand not only high predictive accuracy but also clear and case-specific justifications, both to meet regulatory requirements and to improve communication with applicants.

3 Proposed explainable algorithm

As discussed in the previous section, recent research has increasingly focused on explaining credit risk predictions. Techniques such as SHAP and LIME, commonly used for explaining credit risk predictions, assign importance scores to individual features; however, they do not account for combinations of feature values when explaining a prediction. This limitation highlights the need for approaches that not only assess the contribution of individual features when explaining predictions.

The idea of our algorithm is based on the integration of tree-based predictors with contrast pattern mining algorithms. Unlike traditional explainers for credit risk prediction, which treat the predictor as a black box, our algorithm does not separate the explanation from the prediction process. Instead, it utilizes the predictor's internal structure to generate explanations that reflect its prediction and are consistent with the structure of the training data.

As mentioned previously, credit risk prediction depends on the interaction between multiple features. For instance, a 'high debt-to-income ratio' may only indicate default when combined with 'low job stability'. While existing explanation techniques for credit risk prediction typically assess feature importance individually, our algorithm represents feature interactions using contrast patterns. Contrast patterns provide explanations that reflect combinations of conditions on features, similar to the logical rules used by tree-based predictors when making predictions.

Our choice of tree-based predictors is not only due to their success in state-of-the-art credit risk prediction [38], but also to their natural structural alignment with contrast patterns. This structural compatibility enables a transition

between prediction and explanation. In particular, a decision tree can be interpreted as a hierarchy of decision rules, where each path from the root to a leaf node corresponds to a conjunction of conditions of the form $x_i \wedge v_j$. Similarly, contrast patterns are expressed as conjunctions of conditions on the features.

By employing a contrast pattern mining algorithm compatible with this representation, a direct and consistent correspondence is maintained between the predictor and the generated explanations. This structural alignment mitigates the limitations of approaches such as LIME or SHAP, which can produce explanations that do not fully capture the prediction.

3.1 First stage

The proposed algorithm predicts credit risk using a tree-based predictor and extracts contrast patterns from trees. Because the datasets used for credit risk prediction are imbalanced, and following the results reported in [55], we propose to apply SMOTENC [50, 56] to the training set to address class imbalance.

For prediction, we propose using the FT4cip predictor [57], a tree-based method capable of handling mixed and imbalanced datasets, which has achieved good results with such datasets, making it suitable for credit risk prediction. To construct the explanation, we propose using contrast patterns; therefore, our algorithm focuses on extracting contrast patterns from clusters generated within each class of the dataset for credit risk prediction.

To generate clusters in each dataset for credit risk prediction, we first split the dataset into classes (defaulter and non-defaulter). Within each class, we propose using the Kmodes algorithm [58, 59], and to determine the number of clusters in each class, we propose using Silhouette [60, 61], which has been shown to perform well with mixed-type data. Since we need to construct an explanation for the prediction, contrast patterns are well-suited for this purpose, as they identify features that are frequent in one class and rare or absent in the other.

For this, we use the CPMining pattern miner, introduced in the PBC4cip algorithm [62]. This miner is applied to extract contrast patterns for each cluster of a given class against the other class, ensuring that the resulting patterns are both cluster-specific and class-discriminative.

After mining, we apply a filtering process to remove redundant contrast patterns. Two patterns P_1 and P_2 are compared such that P_1 is considered more general than P_2 if the set of conditions defining P_1 is a subset of those defining P_2. In this case, P_2 is more specific. If both contrast patterns have the same support within the cluster, the more specific pattern is removed because it does not provide additional information. Otherwise, if their supports differ, both contrast patterns are retained, since they describe different subsets of instances. This process reduces redundancy and improves the explainability of the extracted

contrast patterns.

For example, consider the contrast pattern in the German Credit Data dataset [63]:

Credit Duration $\leq$ 15 months **and** Property=real estate **and** Foreign Worker=yes and the more specific contrast pattern

Credit Duration $\leq$ 15 months **and** Property=real estate **and** Foreign Worker=yes **and** Checking Account $\neq$ low

If both contrast patterns cover the same instances within the cluster, the second contrast pattern does not provide additional information, since it introduces an extra condition while describing the same subset of instances. Therefore, only the more general contrast pattern is retained. Additionally, contrast patterns whose support in the opposite class is greater than zero are removed, ensuring that the final set contains only contrast patterns that characterize the cluster while distinguishing it from the other class.

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